Green Mind Metrics: Linking Campus Ecology and Well being

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Overview

10. Green Mind Metrics: Linking Campus Ecology and Well-being

Problem: Many students lack access to quality green spaces despite positive mental-health benefits.

Focus: Map campus greenery through sensors or satellite data; propose a “GreenScore” app or green campus tracker.

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Background and Rationale

Urban university campuses in cities are experiencing rapid physical expansion driven by growing student populations and infrastructure needs. As campuses add buildings, hostels, and roads, the spatial distribution of natural elements such as trees, lawns, gardens, and shaded outdoor areas often changes. At the same time, students in higher education increasingly report stress, burnout, and reduced psychological well-being.

While both campus ecology and student mental well-being are recognized as important, they are usually studied separately. Campuses currently lack a unified, data-driven way to observe how variations in green spaces across different campuses and within the same campus relate to student stress, mood, and perceived well-being. Without such integrated data, students cannot easily identify supportive environments, and institutions cannot prioritize or evaluate environmental interventions for well-being.

This project aims to bridge this gap by combining psychometric survey data and geospatial green-cover data to generate actionable insights and design a data-driven product concept for greener and healthier campuses.

Problem Statement

Large university campuses differ widely in how green spaces such as trees, lawns, gardens, and shaded outdoor areas are distributed, both across different campuses and within different zones of the same campus. At the same time, students across these campuses report differing levels of stress, mood, and psychological well-being. However, campuses currently lack a systematic, data-driven way to measure green cover, compare ecological conditions between campuses, and identify specific locations within a campus that may be associated with more restorative or calming experiences. Without such integrated environmental and well-being data, students and administrators are unable to objectively compare campuses, locate high-value green areas within a campus, or use spatial evidence to guide well-being-focused campus planning, and to understand how campus ecological conditions relate to student well-being. We observe variation in campus green cover and variation in student stress and analyse how it relates. This helps in optimization of campus spaces as active mental health interventions called de-stress green hotspots.

Objectives

The objectives of this project are to:

- Develop a GreenScore and data-driven product concept that helps students and institutions to compare campuses and make informed decisions about campus environments.
- Measure and find the availability and distribution of green spaces ("oases" or the de-stress places) across selected campuses using geospatial data.
- Measure student stress, mood, and well-being using standardized, unbiased survey instruments.
- Analyze relationships between campus greenery and student well-being using statistical and visual methods.

Target Users

- Undergraduate and postgraduate students in university campuses
- Hostellers and day scholars who spend long hours on campus

Data Sources and Collection Plan

It has three data layers:

Layer	What it measures	Why it matters
Psychological	Stress, mood, burnout, calm	Captures human impact
Geospatial	Trees, NDVI, green cover	Captures physical environment
Behavioral	Time spent, usage of green spaces	Captures exposure

1. **Psychometric Data (Student Well-Being)**

A structured survey will be administered to students from multiple campuses. The survey will include:

- Perceived Stress Scale (short form)
- Mood and emotional state indicators
- Time spent in outdoor or green spaces on campus
- Self-reported well-being and academic pressure

Questions are:

- Neutral and non-leading
- Time-bounded (“in the past 1 month”)

- Designed to avoid cognitive biases such as framing or confirmation bias
- The survey was piloted with a small group before wider distribution.

2. Geospatial Green Cover Data

Greenery measured using:

- Satellite imagery
- Sentinel 2 features (NDVI, vegetation indices)
- Socio Urban Features
- LULC(Land Use-Land Cover)

Datasets:

1. **Satellite Image Data**: The satellite image of the specified location (coordinates) is fetched from the Copernicus Sentinel-2 Surface Reflectance (SR) Harmonised dataset. From this 10m resolution raster dataset, a tri-band GeoTIFF with dimensions of 1024×1024 pixels is extracted. This satellite image obtained represents an area of 10.24 square kilometres in a real-life scale with the given coordinates at the centre. For more accurate representation, high-resolution, zoomed-in satellite images of college campuses are generated by stitching together varying grid sizes of Google Maps tiles (e.g., 3x3, 5x5) centered precisely on specific latitude and longitude coordinates. By utilizing a high zoom level (typically 19, approx. 0.5 meters/pixel) and mathematically

shifting the grid's starting point, each image captures fine-grained details of campus infrastructure while ensuring the target institution remains the focal point, with the adjustable tile width determining whether the view is a tight focus on central buildings or a broader perspective of the surrounding area.

2. **Sentinel Data**: Sentinel-2 is part of the European Space Agency's (ESA) Copernicus program, which provides operational environmental monitoring and data services. Sentinel-2 consists of a pair of satellites: Sentinel-2A and Sentinel-2B. Each satellite carries a Multi-Spectral Instrument (MSI) that captures high-resolution images (10 m, 20 m, and 60 m resolution) across 13 spectral bands. Various numerical indices are derived from these bands that contribute to the satellite-based features.

Sl No.	Feature Name	Explanation
1	NDVI	Normalized Difference Vegetation Index - Measures vegetation health
2	Shade	Shade Index - Represents the amount of shaded area
3	NBR	Normalized Burn Ratio - Measures burn severity in vegetation
4	NDUI	Normalized Difference Urban Index - Assesses urban areas
5	GNDVI	Green NDVI - Indicates vegetation greenness and health
6	NDWI	Normalized Difference Water Index - Identifies water bodies
7	NDBI	Normalized Difference Built-up Index - Measures urban built-up areas
8	NMDI	Normalized Multi-band Drought Index - Measures drought conditions
9	NDBSI	Normalized Difference Built-up and Soil Index - Differentiates built-up and soil areas
10	NBaDI	Normalized Bare Soil Index - Detects bare soil areas
11	WVP	Water Vapor Pressure - Measures atmospheric water vapor content
12	AOT	Aerosol Optical Thickness - Indicates air quality and aerosols
13	LST	Land Surface Temperature - Measures the surface temperature of the land
14	SAVI	Soil-Adjusted Vegetation Index - Adjusts for soil brightness
15	UHSI	Urban Heat Stress Index - Measures urban heat stress

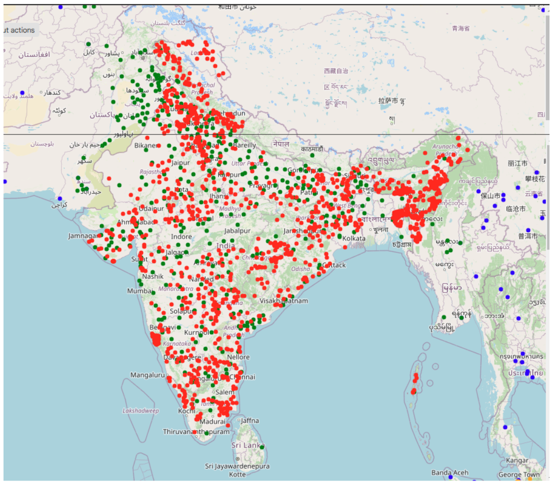
3. **Socio-Urban Data**: The socio-urban features are extracted from sources such as the Copernicus Emergency Management Service, Landsat satellites, Copernicus Sentinel-5P, and National Oceanic and Atmospheric Administration Visible Infrared Imaging Radiometer Suite Day/Night Band (NOAA VIIRS DNB). These datasets provide a comprehensive understanding of urban environments and help in analysing urban expansion, infrastructure development, air quality, and environmental impact in cities

worldwide. These features encompass a variety of urbanisation and environmental indicators.

Sl No.	Feature Name	Explanation
1	Degree of urbanisation	Level of urban development
2	Night light intensity	Reflects urban light pollution
3	Population density per km^2	Number of people per square kilometer
4	Built-up surface	Proportion of urbanised surface area
5	Average gross built height	Average height of buildings
6	Morphological settlement zone	Area based on settlement types
7	Global impervious surface occurrence	Extent of impervious surfaces
8	NO_2 mol per m^2	Nitrogen dioxide concentration per m^2 to indicate air quality
9	pm25 ($\mu g/10m^3$)	Particulate matter concentration PM2.5 used as air quality indicator

4. **LULC Mask Data**: The sample LULC masks required to train the semantic segmentation model are obtained from the global LULC with Sentinel-2 and deep learning Environmental Systems Research Institute (ESRI) dataset. This dataset comes from a study that used Sentinel-2 imagery in combination with deep learning models to classify global LULC. The dataset classifies the Earth's surface into nine categories, like forests, water bodies, urban areas, agriculture, and more. The LULC masks corresponding to the

4771 locations are fetched from this raster dataset and are used as a single band Geographic Tagged Image File Format (GeoTIFF) file with 9 classes for modelling.



Sl No.	Feature Name	Explanation
1	Water ratio	Proportion of water bodies in the area
2	Trees ratio	Proportion of tree cover in the area
3	Flooded Vegetation ratio	Proportion of flooded vegetation in the area
4	Crops ratio	Proportion of crop-covered land
5	Built Area ratio	Proportion of urban built-up areas
6	Bare Ground ratio	Proportion of bare or uncovered ground
7	Snow/Ice ratio	Proportion of snow or ice-covered areas
8	Clouds ratio	Proportion of clouds present in the area
9	Rangeland ratio	Proportion of rangeland or pasture areas

For each campus:

- Percentage green cover
- Tree density (approximate)
- Distribution of green areas (centralized vs scattered)

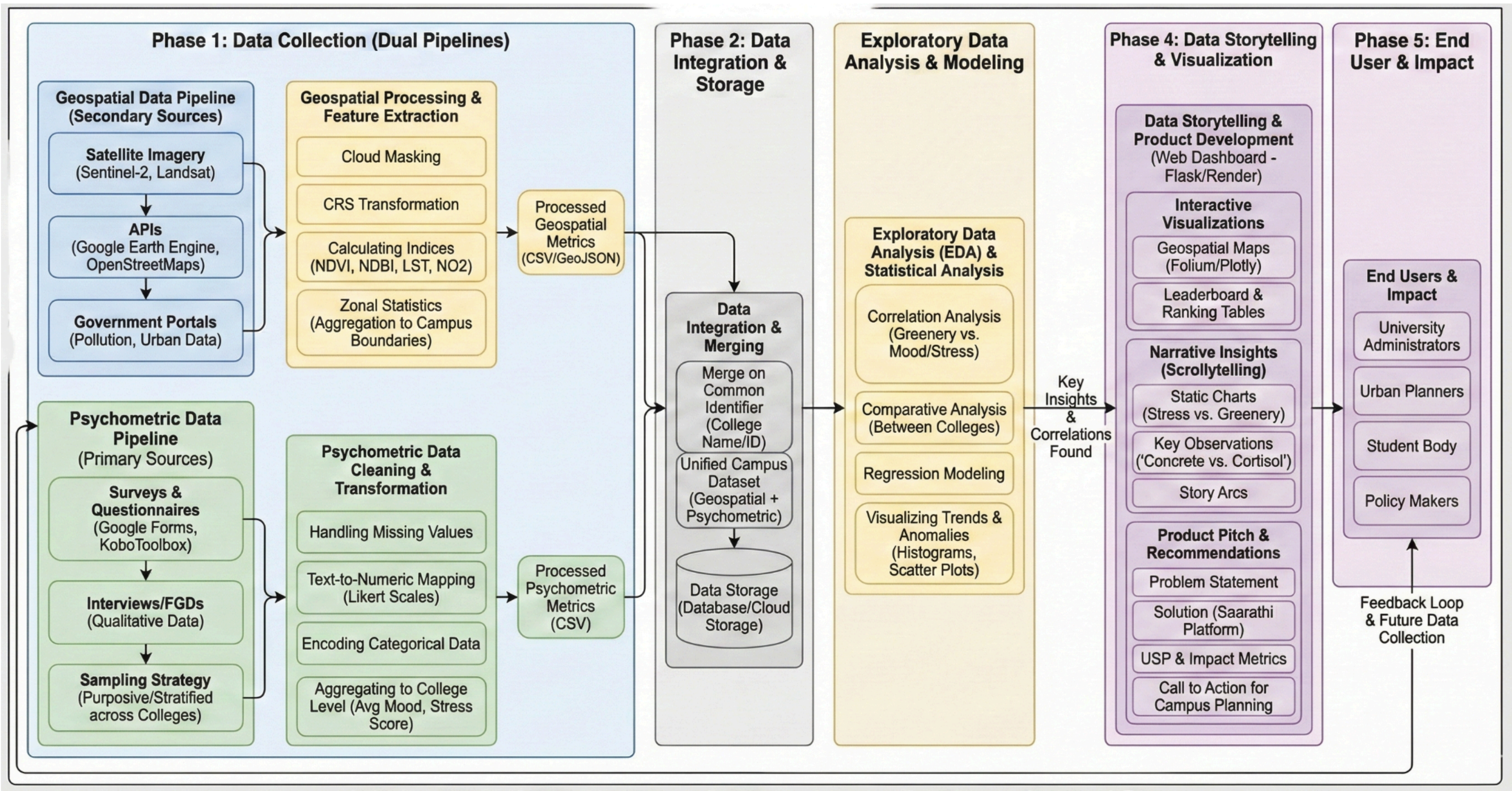
Sampling Plan

Population: Students enrolled in universities

Sampling Method: Stratified convenience sampling across multiple campuses

Target Sample Size: 80–150 students

Data Processing and Analysis Plan



1. Overview

This project investigates the correlation between campus environments ("Green Score") and student well-being ("Psychometric Score") across 42 Indian colleges. The methodology employs a dual-pipeline approach, integrating satellite-derived geospatial data with primary psychometric survey data to drive a data-driven product pitch for "Green Health" audits in educational institutions.

2. Phase 1: Data Collection (Dual Pipelines)

- **Pipeline A: Geospatial Data (Secondary Sources)**
  - **Data Sources:**
    - **Google Earth Engine (GEE):** Primary platform for satellite data acquisition and processing.
    - **Copernicus Sentinel-2:** Multispectral imagery (13 bands, 10m resolution) used to derive vegetation and water indices.
    - **Socio-Urban Data:** Extracted from multiple sources within GEE, including Landsat (historical context), Sentinel-5P (pollution metrics like NO2), and NOAA VIIRS (Night-time lights for urbanization intensity).
    - **ESRI Global LULC Dataset:** Ground truth land cover masks used for training the custom segmentation model.
  - **Processing:**
    - **Custom U-Net Model:** A deep learning semantic segmentation model was trained on the ESRI dataset to perform custom Land Use Land Cover (LULC) classification on the campus images.

    - **Spectral Indices:** Calculation of key environmental metrics:
        - **NDVI & GNDVI:** Vegetation health and density.
        - **NDBI:** Normalized Difference Built-up Index (urban density).
        - **LST:** Land Surface Temperature (heat islands).
    - **Zonal Statistics:** Aggregation of pixel-level data to the "College Campus" boundary level to create a single row of metrics per institution.
  - **Output:** `geospatial_college_data_full.csv` containing normalized environmental scores for each college.
- **Pipeline B: Psychometric Data (Primary Source)**
  - **Data Source:**
    - **Google Forms Survey:** Floated across 42 colleges in India.
    - **Respondents:** Students provided self-reported data on stress, mood, focus, and perceived campus greenness.
  - **Processing:**
    - **Cleaning:** Handling missing values and standardizing college names to match the geospatial dataset.
    - **Quantification:**
        - **Likert Scale Mapping:** Converting text responses (e.g., "Always", "Rarely") into numeric scores (1-5 or 0-100).
        - **Encoding:** Categorical variables (Gender, Residence Type) were encoded for analysis.
    - **Aggregation:** Individual student responses were averaged by college to create a "Campus Psychometric Profile" (Mean Mood, Mean Stress).

  - **Output:** `psychometric_college_data.csv` containing aggregated well-being metrics per college.

3. Phase 2: Data Integration

- **Merging Strategy:** The two datasets were merged using the unique identifier `Short Name` / `College Name`.
- **Result:** A unified Master Dataset where every row represents a college, containing both its physical environmental stats (NDVI, Pollution) and its human capital stats (Student Mood, Stress).
- **Storage:** The processed data is stored in CSV format, ready for the analysis engine.

4. Phase 3: Exploratory Data Analysis (EDA) & Modeling

- **Correlation Analysis:** Investigating the relationship between "Green" metrics (NDVI) and "Mind" metrics (Stress/Mood).
  - *Key Insight:* Strong negative correlation found between high concrete density and student stress levels.
- **Comparative Analysis:** Benchmarking "Green Campuses" vs. "Non-Green Campuses" to observe differences in average mood scores.
- **Regression Modeling:** Using environmental factors (PM2.5, Green Cover) to predict potential student stress levels.

- **Visualizations:** Histograms of mood distribution, scatter plots of Stress vs. Greenery, and box plots for college-wise comparisons.

5. Phase 4: Data Storytelling & Visualization (The Dashboard)

https://green-mind-metrics.onrender.com/

- **Web Framework:** Built using **Flask (Python)** for the backend and **Alpine.js/HTML/CSS** for the frontend, hosted on **Render**.
- **Interactive Visualizations:**
  - **Geospatial Maps:** Folium/Plotly maps displaying the 42 colleges, color-coded by their "Green Score".
  - **Leaderboard:** A dynamic ranking table sorting colleges by their calculated "SustainScore".
- **Narrative Scrollytelling:** A dedicated `insights.html` page uses a "Scrollytelling" design to walk the user through the findings:
  - *Section 1:* "Concrete vs. Cortisol" – Visualizing the link between built-up area and stress.
  - *Section 2:* "The Green Dopamine Hit" – Showing mood improvements in green zones.
  - *Section 3:* "Focus Analysis" – How visual greenery aids academic concentration.

6. Phase 5: Product Pitch & Impact

- **Problem Statement:** Campuses are managing resources (water/electricity) but ignoring the "Psychological Green Score" of their environments.
- **Solution:** An AI-driven "Green Audit" platform.
    - *Feature 1:* Automated Satellite Audits (No sensors needed).
    - *Feature 2:* Student Feedback Heatmaps.
    - *Feature 3:* Actionable planting recommendations for stress reduction.
- **End Users:** University Administrators, Urban Planners, Student Welfare Bodies.
- **Impact:** Validated improvement in student focus (80% reported) and reduction in stress variance (32%) in greener campuses.

Exploratory Data Analysis

The following analysis examines the human dimension of the "Green Mind Metrics" study, utilizing survey data from students across multiple institutions to understand the intersection of campus ecology and mental well-being.

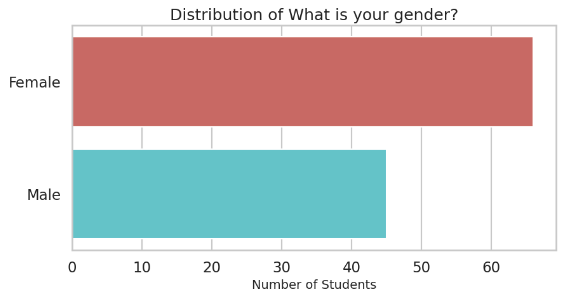
Demographics

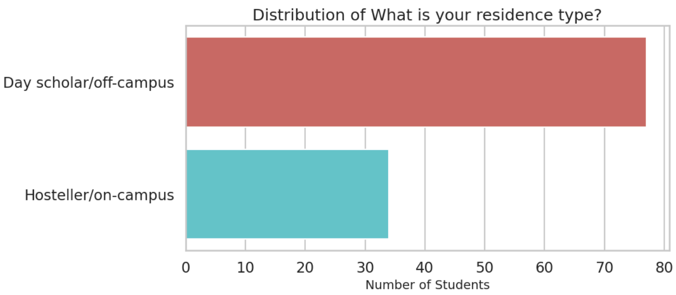
Establishes the profile of the study group by identifying their gender distribution, residential status, and geographic origins to ensure the sample is representative.

1.Geographic Distribution

The geographic mapping shows a strong concentration of respondents from South India, particularly within the Bengaluru and Chennai academic hubs. While the survey reached students across the country, this regional density suggests the data is highly reflective of students within India's major urban and technical educational centers.

2. Gender Representation and Residence Type



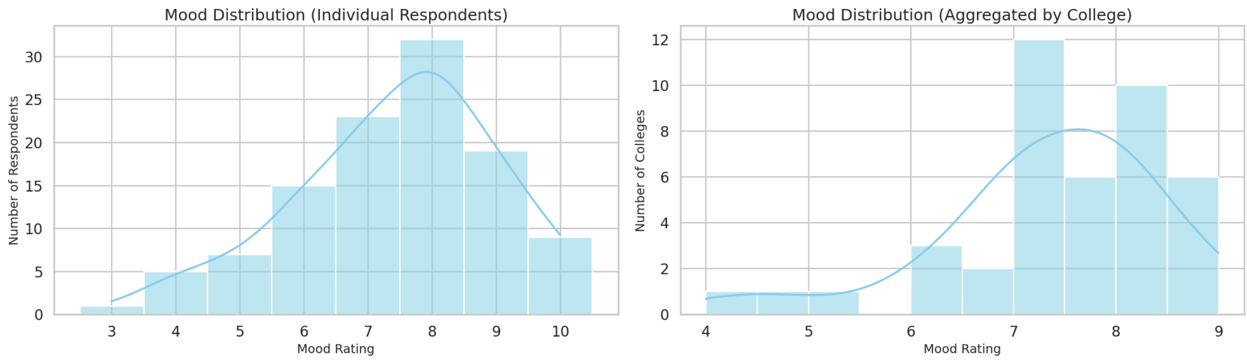


The survey cohort represents a diverse student population with a clear **female majority**, comprising approximately **60%** of participants (66 respondents) compared to **40%** identifying as male (45 respondents). This balanced foundation is complemented by a distinct residential profile, where **Day scholars** make up the vast majority. With **77 students** living off-campus more than double the **34 hostellers** the data primarily reflects the experiences of individuals managing independent commuting and off-campus lifestyles. Together, these metrics provide a robust demographic lens for analyzing how gender and residential stability influence psychometric trends and environmental interactions.

Students' Mood, Stress & Focus

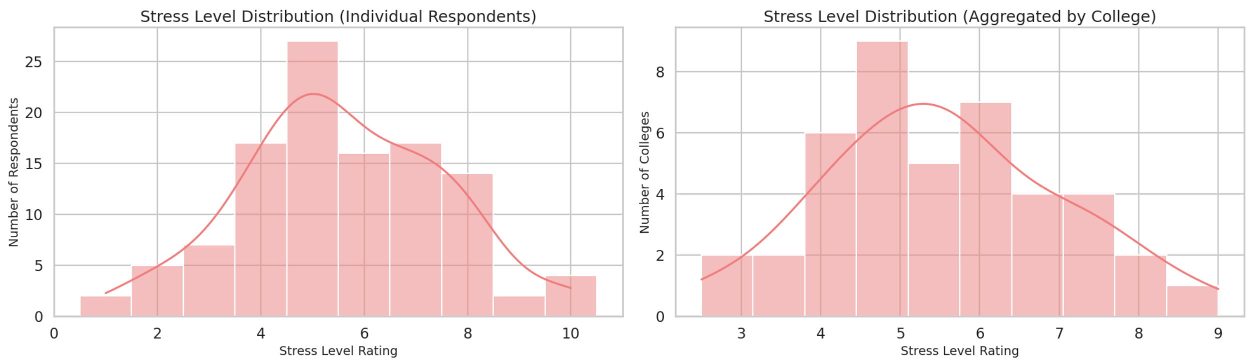
Analyzes the current mental and emotional baseline of students, identifying how their general well-being correlates with internal factors.

1. Mood and Well-being Distribution



The survey indicates a generally positive emotional baseline among students. Individual mood ratings peak at **8 out of 10**, with a broad distribution suggesting that most students feel positively on a day-to-day basis. When aggregated by institution, the data remains consistent, showing that the majority of colleges maintain a "healthy" average mood rating between **7 and 8**.

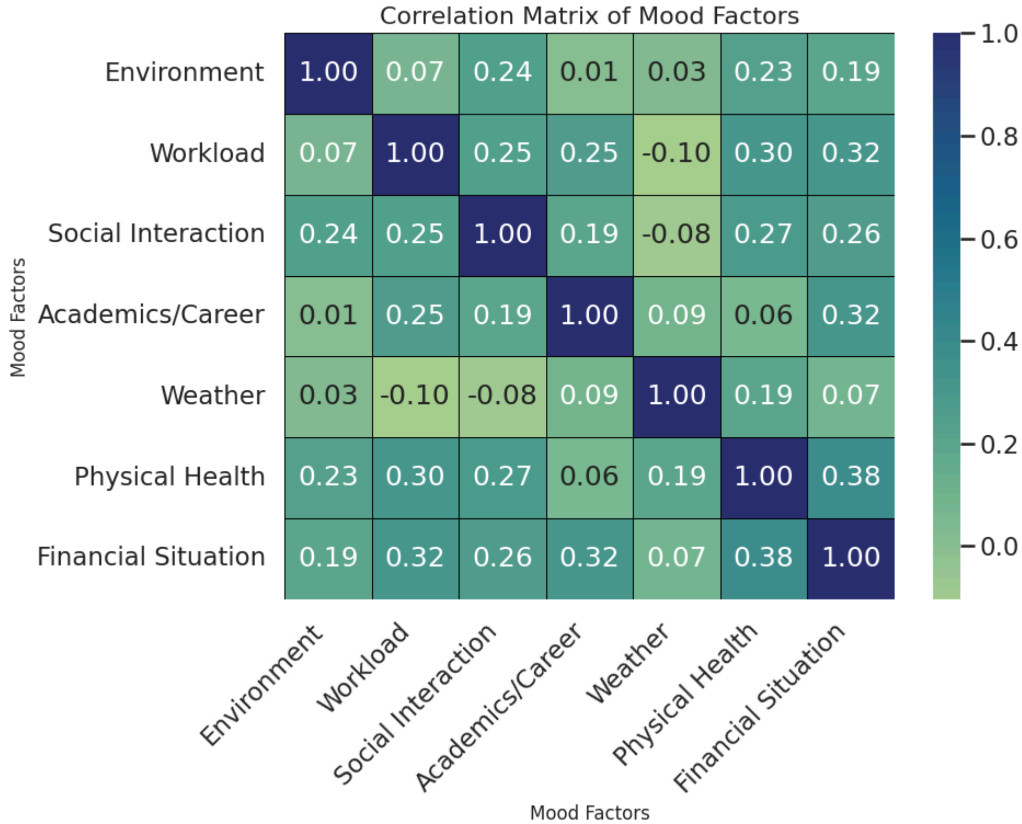
2. Stress Level Analysis



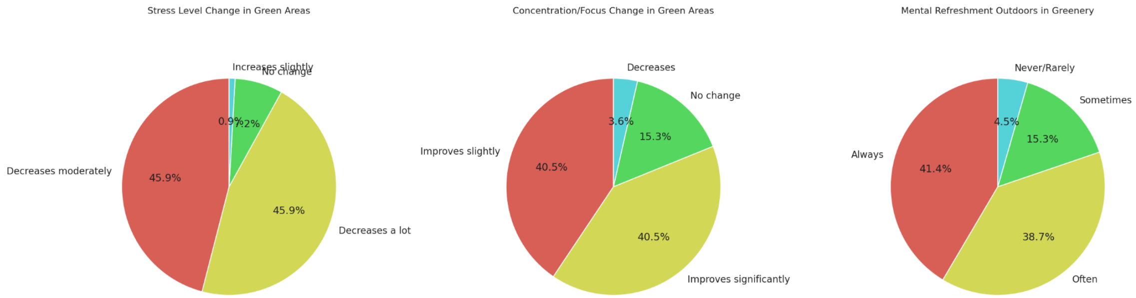
In contrast to the high mood ratings, stress levels follow a more centered, bell-shaped distribution. The primary concentration of respondents reports a stress level of **5 out of 10**, indicating moderate academic and personal pressure. The data suggests that while students are generally happy, they are simultaneously managing a consistent and significant amount of stress.

3. Key Drivers of Mood (Correlation)

According to the correlation matrix, student mood is most significantly influenced by **Financial Situation (0.38)** and **Physical Health (0.38)**. Interestingly, factors like **Weather (0.07)** and **Environment (0.23)** have a much lower direct correlation with general mood, suggesting that internal and stability-based factors outweigh external atmospheric conditions in determining well-being.

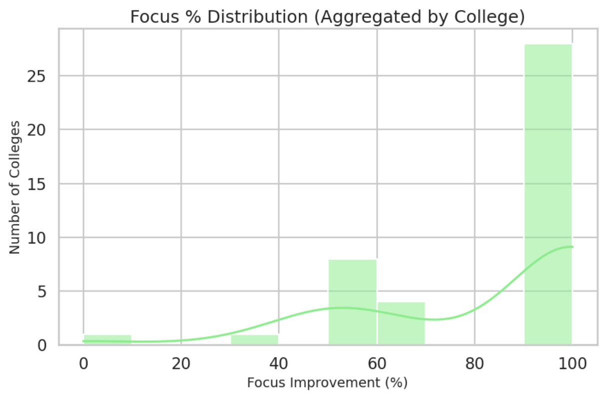


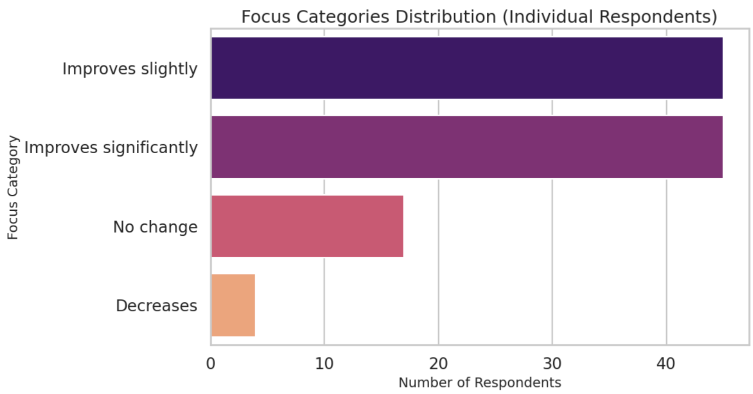
4. Impact of Green Spaces on Mental Health



Nature acts as a powerful intervention for student well-being. **91.8%** of participants report a reduction in stress when spending time in green areas, with nearly half (45.9%) stating their stress "decreases a lot." Additionally, **80.1%** of students feel mentally refreshed "Always" or "Often" when outdoors in greenery, highlighting a nearly universal positive response to natural environments.

5. Focus and Concentration Enhancement



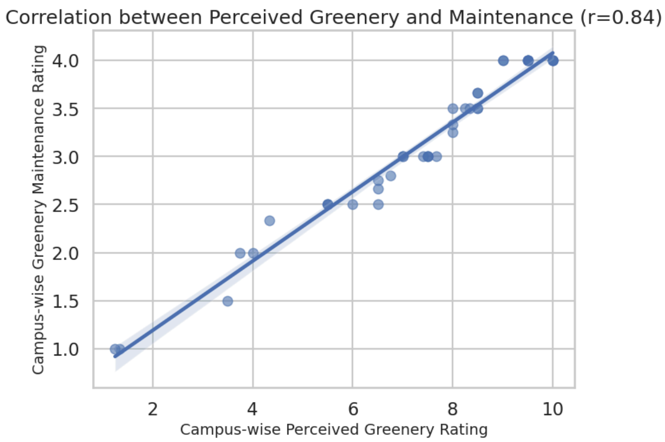


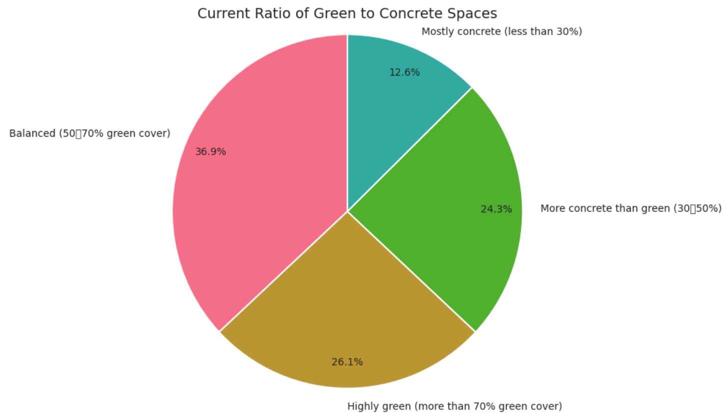
Green environments significantly improve academic performance through better focus. **81%** of respondents reported that their concentration improves either "significantly" or "slightly" in green settings compared to non-green areas. The data shows a massive concentration of focus improvement nearing **100%** across many colleges, reinforcing the value of biophilic design on university campuses.

Campus & Green Spaces

Investigates how students physically interact with their environment, measuring the actual time spent in nature versus built infrastructure and the perceived quality of those spaces and measure of the psychological impact of natural environments on Students.

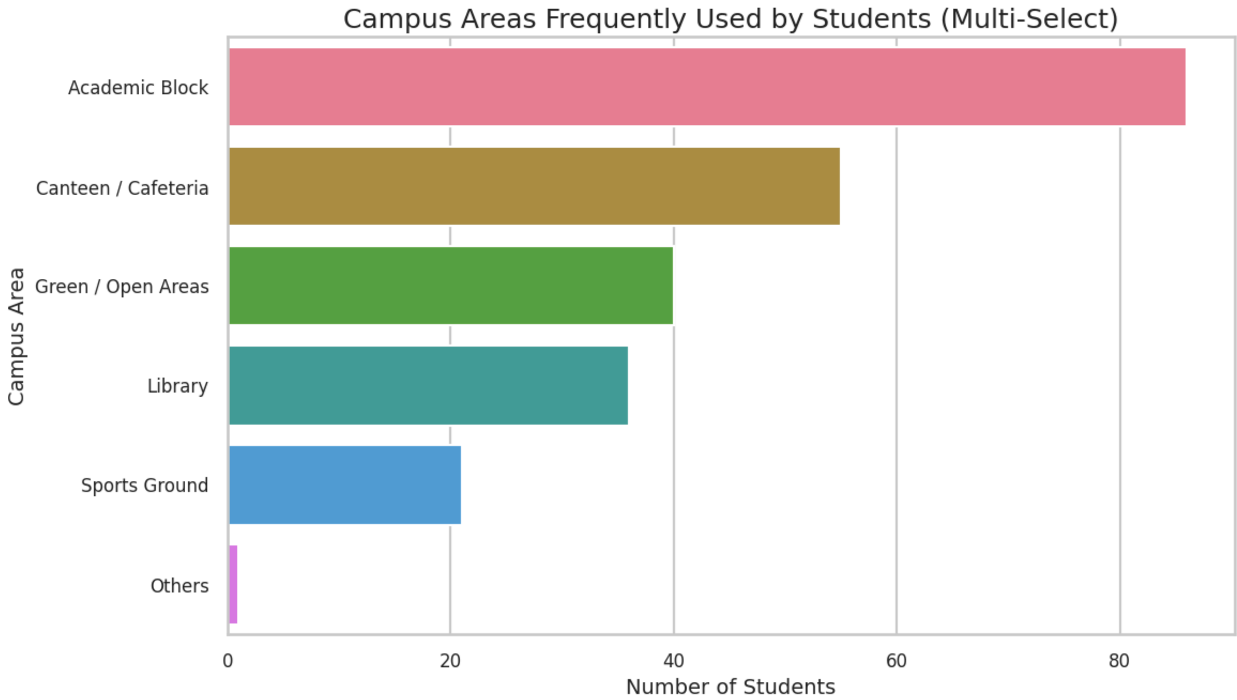
1. Campus Greenery Composition





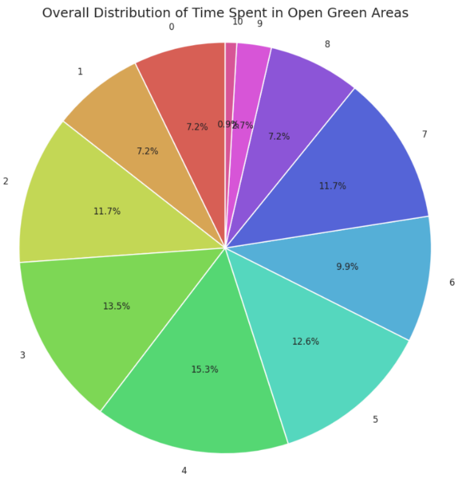
The architectural makeup of participating campuses shows a leaning toward balanced environments, with **36.9%** of students reporting a **50-70% green cover**. While approximately **26%** of students enjoy highly green campuses (over 70% cover), a significant **36%** are situated in environments that are predominantly concrete. This variance provides a strong baseline to compare how different levels of "green exposure" affect student well-being.

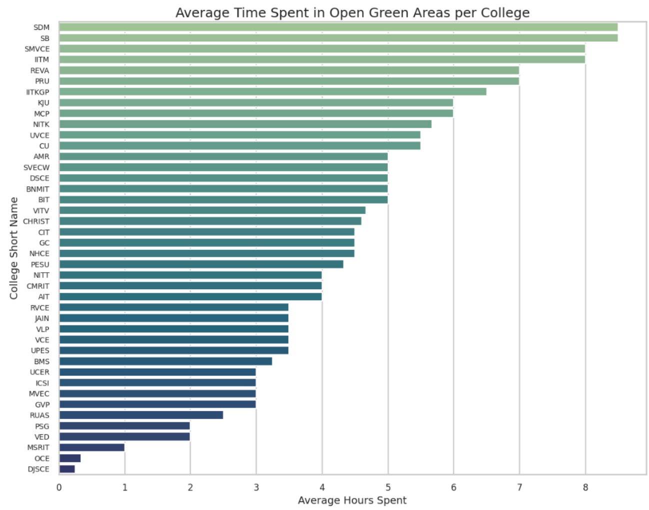
2. Utilization of Campus Areas



Despite the availability of green spaces, students are primarily tethered to built infrastructure for their daily activities. **Academic blocks** are the most frequently used areas (over 80 respondents), followed by **Canteens**. **Green and Open Areas** rank third in frequency, used by approximately 40 students, followed closely by libraries. This suggests that while green spaces are valued, they are currently secondary to functional academic and social hubs.

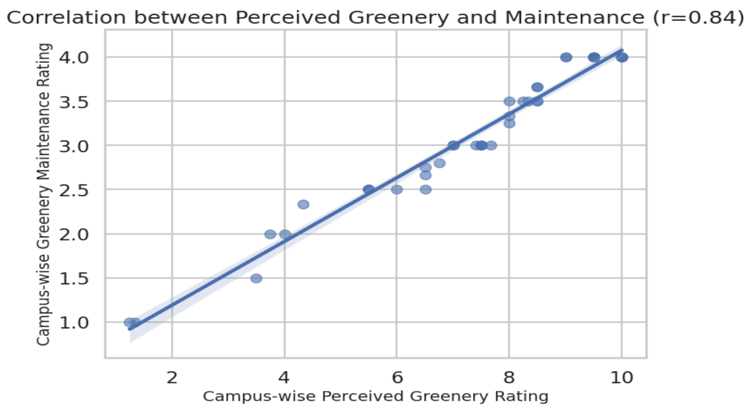
3. Time Spent in Nature





Daily engagement with the outdoors varies significantly across the cohort. The largest single group of students (**15.3%**) spends roughly **4 hours** a day in open green areas. However, the distribution is broad; while some elite campuses see averages as high as **8 hours**, a substantial portion of the student body spends **2 hours or less** in nature during a typical day.

4. Perceived Quality and Maintenance

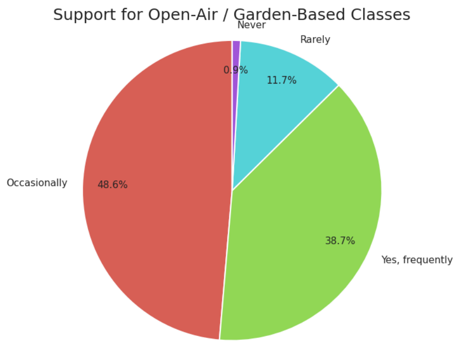


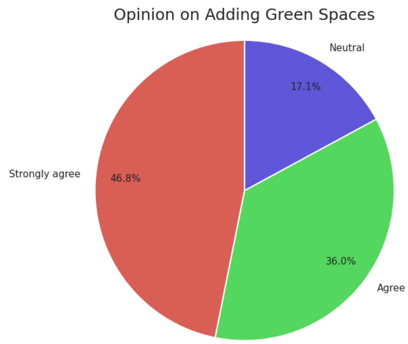
There is a powerful linear relationship between how green a campus is perceived to be and how well its greenery is maintained. Students do not just value the quantity of green space, but the **quality** of its upkeep. Higher maintenance ratings directly correlate with higher perceived greenery, indicating that investment in landscaping is a key driver of a student's positive environmental perception.

Reimagining Campus Spaces

Explores future possibilities and student opinions on integrating biophilic design, such as open-air classrooms, to permanently improve the campus experience.

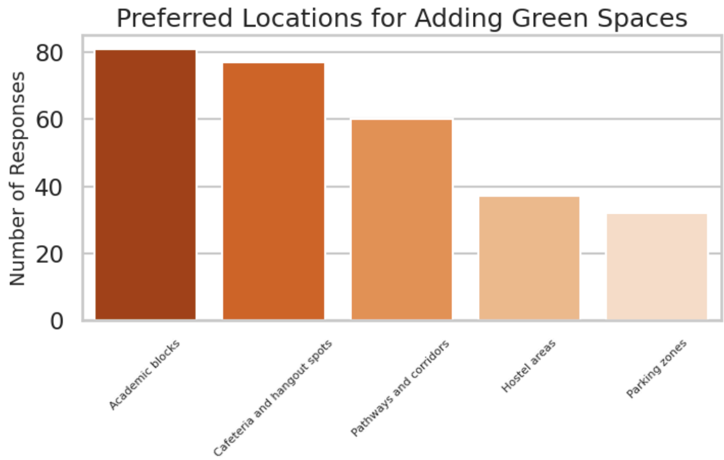
1. Support for Biophilic Learning





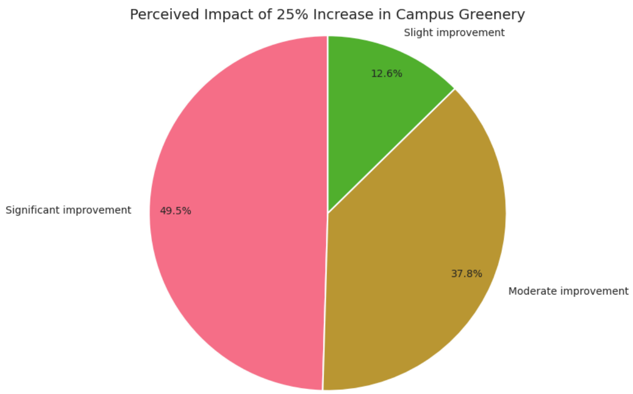
There is overwhelming student support for integrating natural elements into the academic curriculum. **87.3%** of respondents support the idea of **open-air or garden-based classes**, with 38.7% favoring them "frequently" and 48.6% "occasionally." This indicates a strong desire to move beyond traditional four-walled classrooms into more integrated, natural learning environments.

2. Strategic Priorities for Greenery



Students have identified high-traffic functional zones as the most critical areas for additional green interventions. **Academic blocks** and **Cafeterias/Hangout spots** are the top priorities, receiving the highest number of requests for increased greenery. This suggests that students want "greenery where they are," rather than isolated parks, focusing on the spaces where they study and socialize most.

3. Perceived Impact of Future Greenery



The community is nearly unanimous in the belief that more green spaces would enhance student well-being, with **82.8%** in agreement (46.8% strongly agreeing). When asked specifically about the impact of a **25% increase in campus greenery**, almost **90%** of participants anticipated a positive result, with 49.5% expecting a "significant improvement" in their mental health.

Summary of Sentiment & Vision

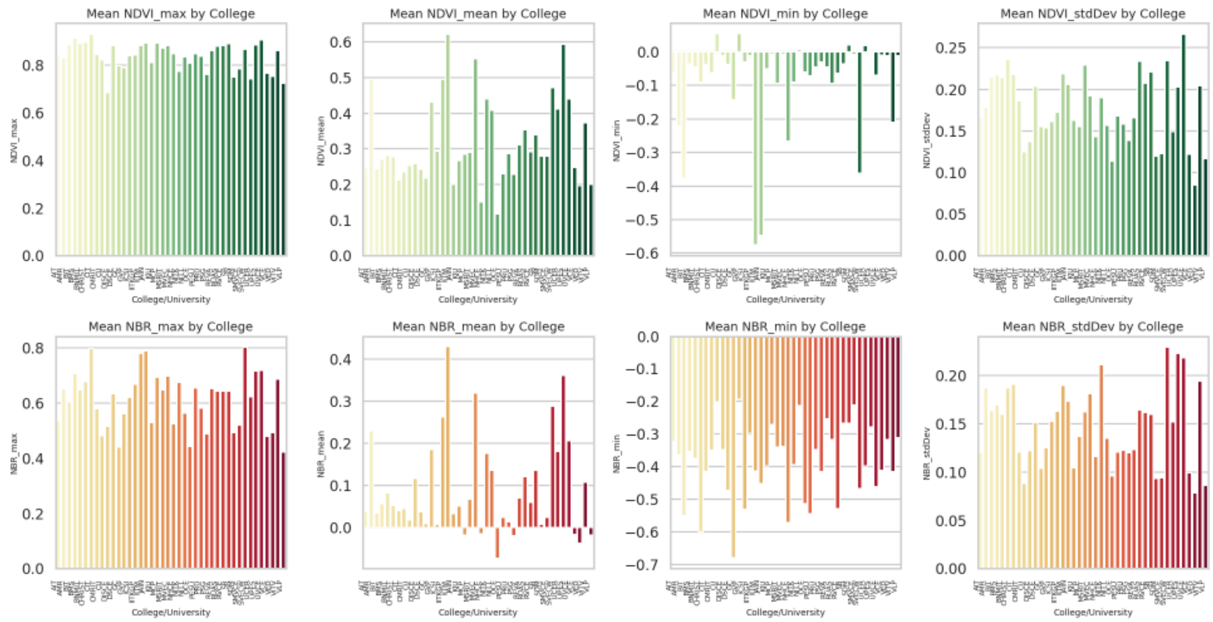
The data reflects a student body that is highly conscious of the psychological benefits of their physical environment. The vision for a reimagined campus is one that prioritizes **high-quality maintenance** and **functional integration**. By moving greenery into academic and social corridors, institutions can directly address the moderate stress levels and high focus requirements identified in earlier sections of this analysis.

Comparative & Correlation Analysis

Integrated Geospatial and Urban Environmental Analysis

This section correlates high-resolution satellite vegetation indices with on-ground student perceptions to validate the ecological quality of campus landscapes.

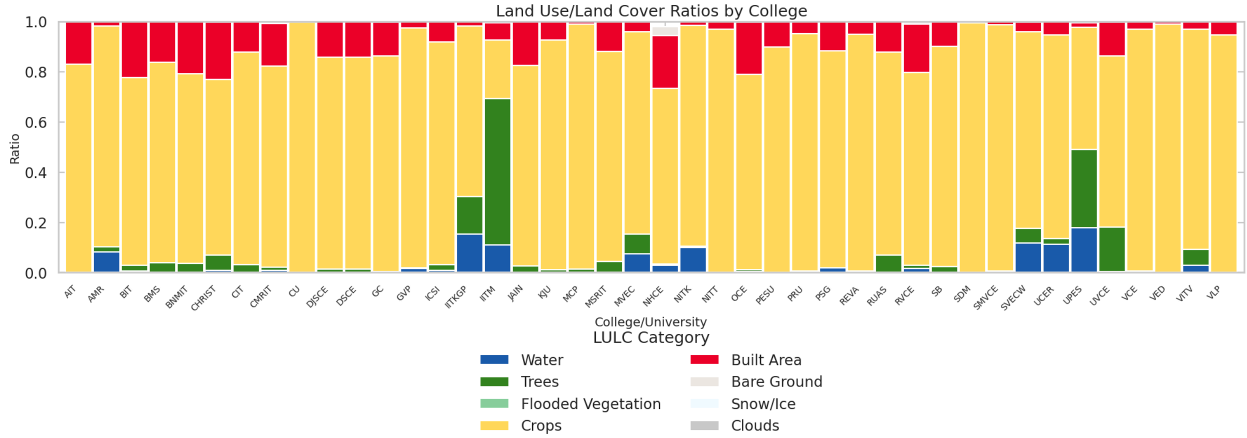
1. Sentinel Environmental Indices by College



Observation: Satellite mapping reveals a sharp contrast in environmental signatures across the colleges studied. Campuses are clearly segmented into "Green Havens," characterized by high NDVI (Normalized Difference Vegetation Index) and NDWI (Normalized Difference Water Index), and "Urban Hubs," which exhibit elevated NDBI (Normalized Difference Built-up Index) levels.

Insight: These metrics prove that the physical differences between campuses are not merely aesthetic but are scientifically measurable ecological realities. Institutions with higher water and vegetation indices demonstrate superior environmental resilience, providing an objective baseline that distinguishes greener campuses from those defined by heavy structural development.

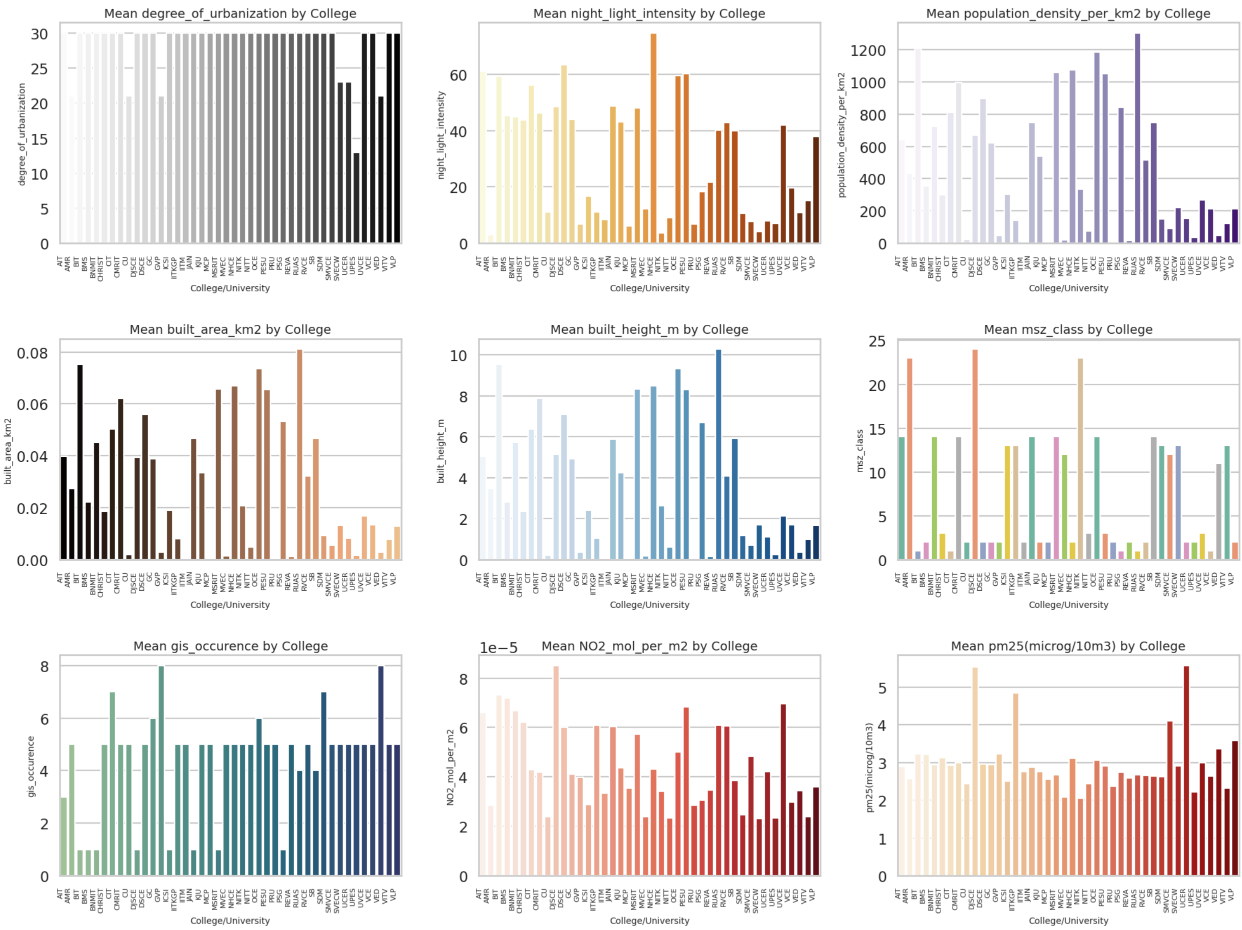
2. Land Use/Land Cover (LULC) Ratios by College



Observation: The data reveals a clear "Ecological Gap" where most campuses are dominated by agricultural Crops (yellow segments), often exceeding an 80% ratio, while high-density Tree canopy (dark green) and Water bodies (blue) remain remarkably scarce. Only a few benchmark institutions like IITM and UPES demonstrate significant natural diversity with substantial tree cover ratios.

Insight: This relates to your project's focus on "quality green spaces". While agricultural openness may be perceived as "green," it often lacks the shading and complex biophilic structures scientifically proven to lower stress. The GreenScore app can use this distinction to explain why "viewing crops" may not offer the same restorative benefits as a shaded forest canopy.

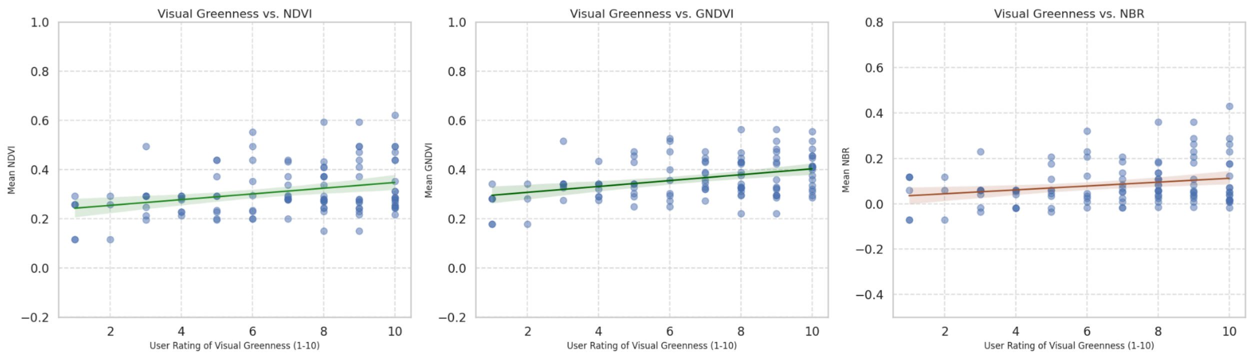
3. Socio-Urban & Pollution Metrics by College



Observation: Analysis shows a clear "Urban Cluster" where high population density, night-light intensity, and building height are directly linked to elevated concentrations of pollutants like NO_2 and PM2.5. More "active" and dense campuses consistently exhibit poorer air quality metrics.

Insight: Urbanization brings a "double burden" to campus life, these environments are not only psychologically taxing due to high density, but they are also chemically distinct. This suggests that dense, built-up campuses pose a higher physical health risk to students, reinforcing the need for green buffers to mitigate the effects of urban pollution.

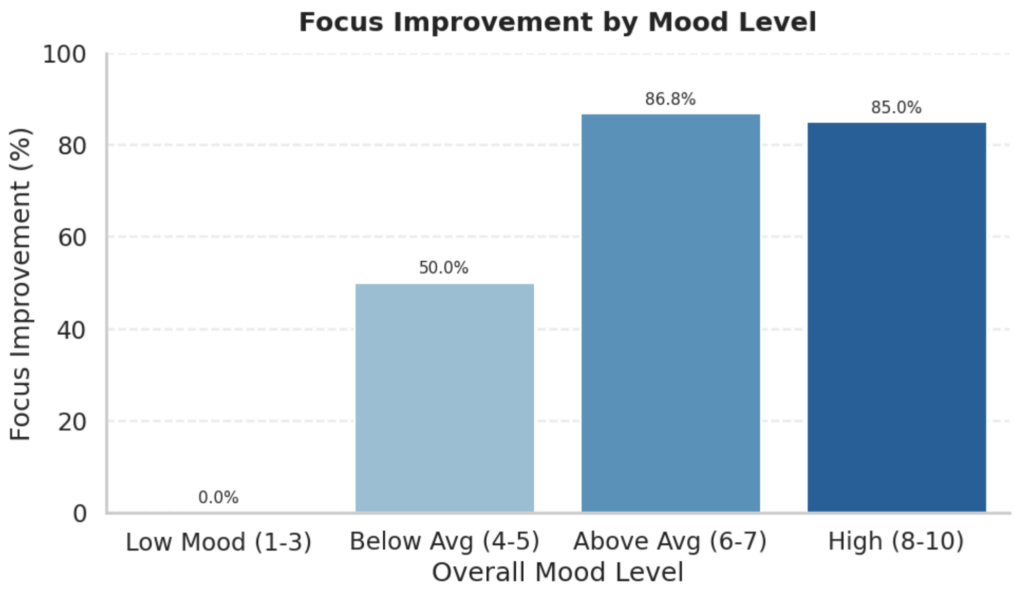
4. Visual Greenness vs. NDVI / GNDVI / NBR



Observation: There is a strong, near-linear correlation between objective satellite indices (like NDVI and GNDVI) and the subjective ratings provided by students regarding their campus's visual greenness. As the measured density of healthy vegetation increases from orbit, student satisfaction with the environment rises in lockstep. **Insight:** These findings bridge the gap between data science and psychology, confirming that students are highly accurate sensors of their ecological surroundings. It validates that investing in high-quality greenery is directly perceived and appreciated by the student body, making it a reliable lever for improving student satisfaction.

Mood Analysis

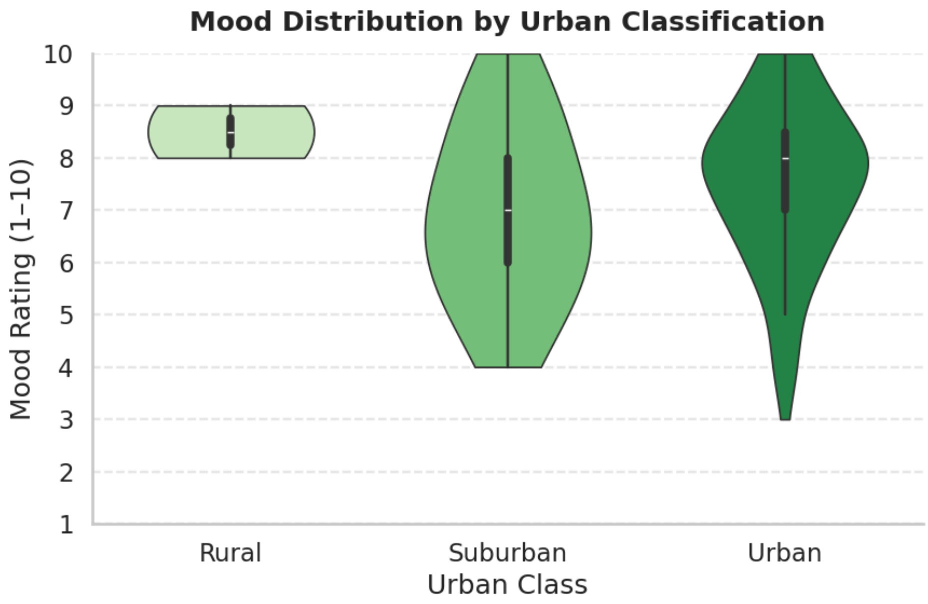
1. Focus Improvement by Mood Level



Observation: This bar chart highlights a "threshold effect" where cognitive performance is directly dependent on emotional well-being. Students experiencing a "Low Mood" (1–3) report 0.0% focus improvement, while those in the "Above Average" (6–7) and "High" (8–10) categories report massive gains of 86.8% and 85.0% respectively.

Insight: These findings indicate that positive mood is a prerequisite for academic focus. The jump from 50% to over 85% improvement as mood moves from "Below Average" to "Above Average" suggests that institutional efforts to boost student happiness perhaps through the green spaces mentioned above can directly unlock significantly higher levels of academic productivity.

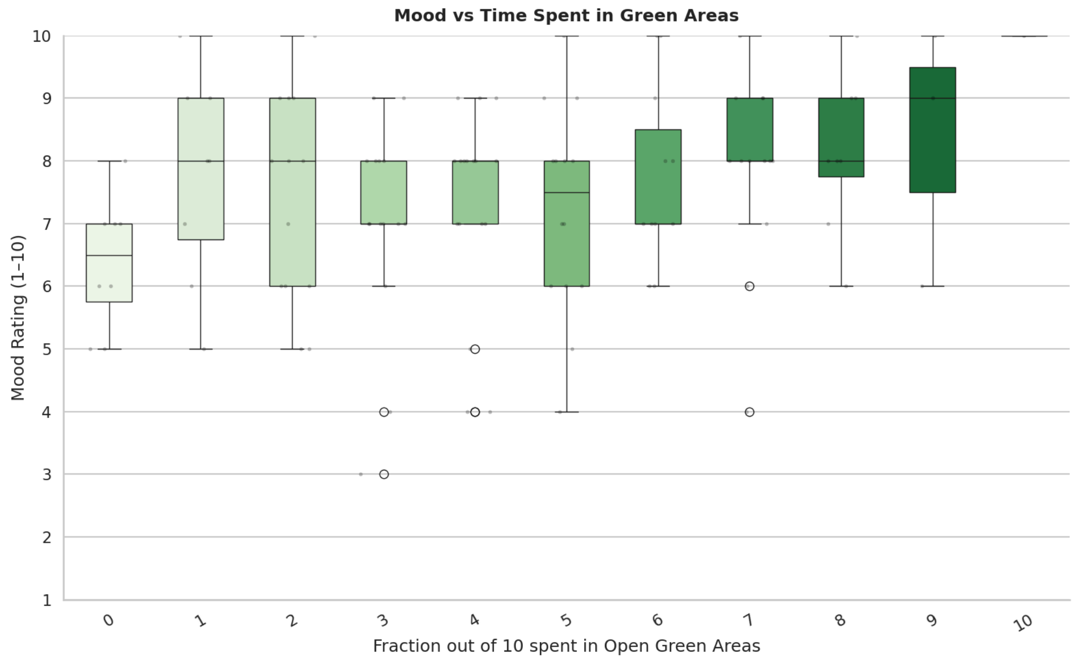
2. Mood Distribution by Urban Classification



Observation: This violin plot illustrates the stability and spread of emotional well-being across different living environments. Students from Rural backgrounds show the most consistent and concentrated mood distribution, primarily clustered between 8 and 9. In contrast, Suburban and Urban students display a much wider "violin" shape, indicating higher variability and significantly more frequent occurrences of lower mood ratings (dropping as low as 3 or 4).

Insight: These findings suggest that while urban centers offer more resources, they also introduce higher emotional volatility for students. The rural environment appears to provide a natural "emotional floor," whereas suburban and urban living may contribute to a wider range of stressors that disrupt emotional stability.

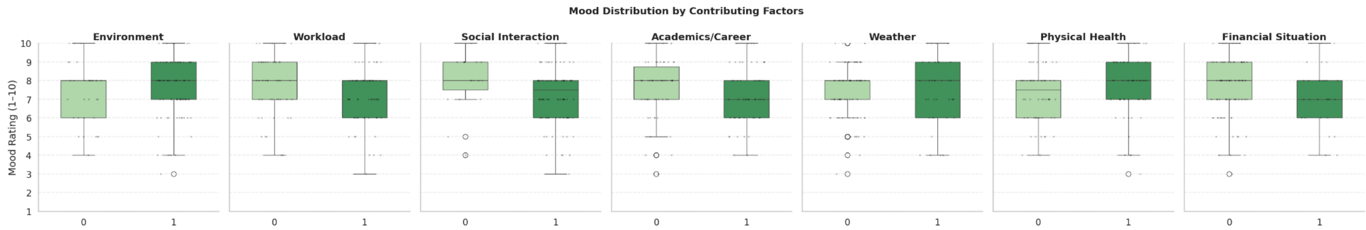
3. Mood vs. Time Spent in Green Areas



Observation: This boxplot demonstrates a strong positive relationship between the amount of time students spend in nature and their emotional state. Students with the lowest exposure to green spaces (0–2 scale) report median mood ratings of approximately 6.5, whereas those who spend nearly all their time in these areas (9–10 scale) show a significant rise to median scores between 8.5 and 10.

Insight: The data suggests that nature acts as a consistent "emotional elevator". As the fraction of time spent in green areas increases, the "floor" for student mood also rises, indicating that green spaces provide a reliable buffer that prevents mood from dropping into negative territories.

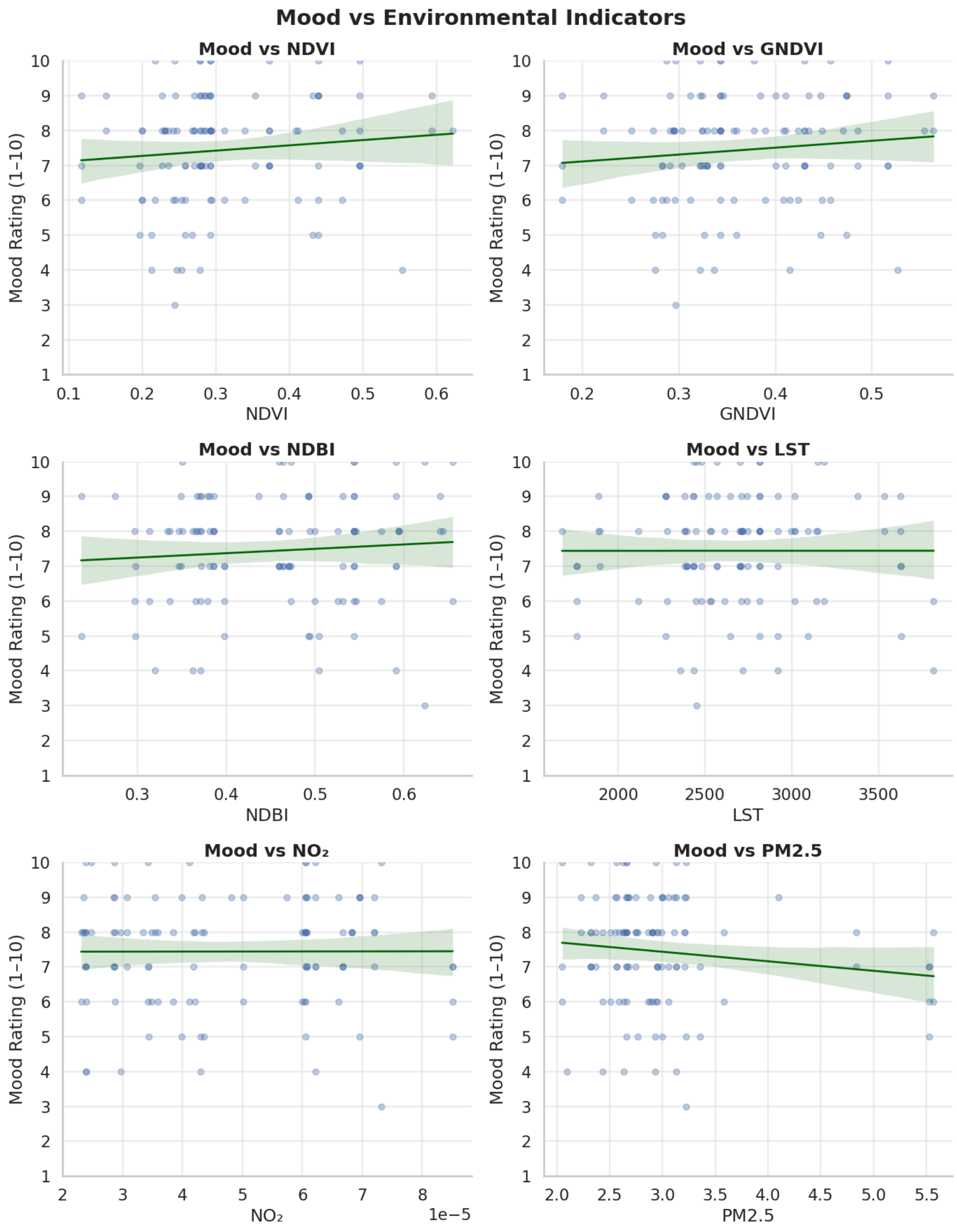
4. Mood Distribution by Contributing Factors



Observation: This multi-panel boxplot identifies which specific life factors most effectively "lift" or "drag" student mood. Environment, Weather, and Physical Health show a distinct upward shift in median mood (from approximately 7 to 8) when these factors are positive (marked as '1'). Conversely, high Workload and Financial Situation pressures show a wider distribution of lower mood scores, dragging the lower quartile of responses down to 6 or below.

Insight: This data confirms that well-being is a multifaceted "balance sheet". While external environment and health provide a clear emotional boost, internal pressures like workload and finances are more likely to cause severe "mood dips". This emphasizes that environmental interventions (like green spaces) are most effective when paired with academic workload management to protect the student's emotional baseline.

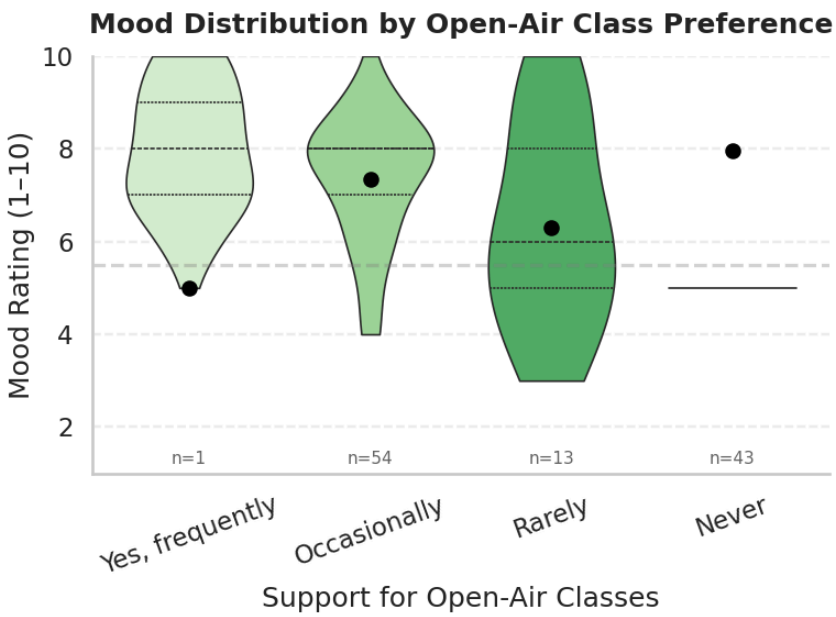
5. Mood vs. Environmental Indicators


Mood vs Environmental Indicators
Mood vs NDVI
Mood vs GNDVI
Mood vs NDBI
Mood vs LST
Mood vs NO₂
Mood vs PM2.5
Mood Rating (1–10)
NDVI
GNDVI
NDBI
LST
NO₂
1e−5
PM2.5

Observation: Regression analysis shows that objective environmental quality directly correlates with student happiness. Higher vegetation indices, such as NDVI and GNDVI, show a positive relationship with mood ratings. In contrast, pollutants like PM2.5 show a clear negative trend, where increased air pollution consistently drags down the overall mood of the student body.

Insight: This data proves that student well-being is not just a psychological state but a biological response to their physical surroundings. The "Mood-Environment Link" confirms that greener campuses are literally happier campuses, while air quality serves as a silent but significant detractor of student mental health.

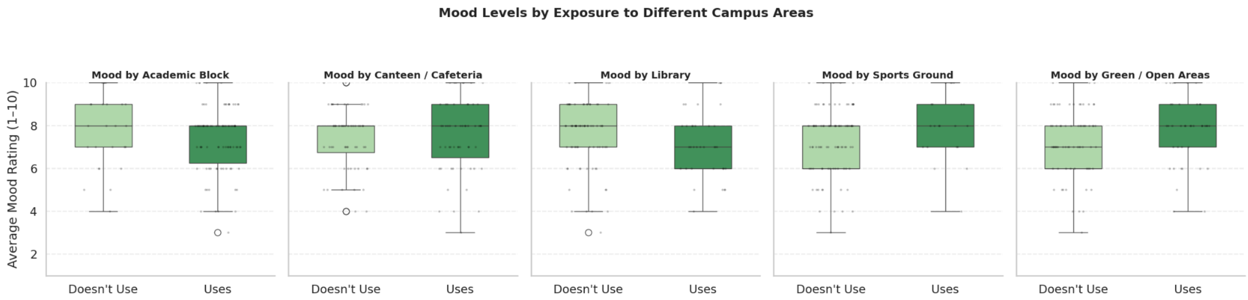
6. Mood Distribution by Open-Air Class Preference



Observation: The violin plot indicates a correlation between the desire for nature-integrated learning and overall happiness. Students who favor open-air classes "Frequently" or "Occasionally" exhibit higher median mood ratings compared to those who "Rarely" or "Never" prefer them. Additionally, the group that "Frequently" supports these classes shows the most concentrated and stable high-mood distribution.

Insight: This data suggests that students who are already more emotionally positive have a higher affinity for natural learning environments. Alternatively, it implies that the psychological profile of a high-mood student is more open to biophilic (nature-based) educational settings, whereas lower mood states may correlate with a more traditional or restricted classroom preference.

7. Mood Levels by Exposure to Different Campus Areas

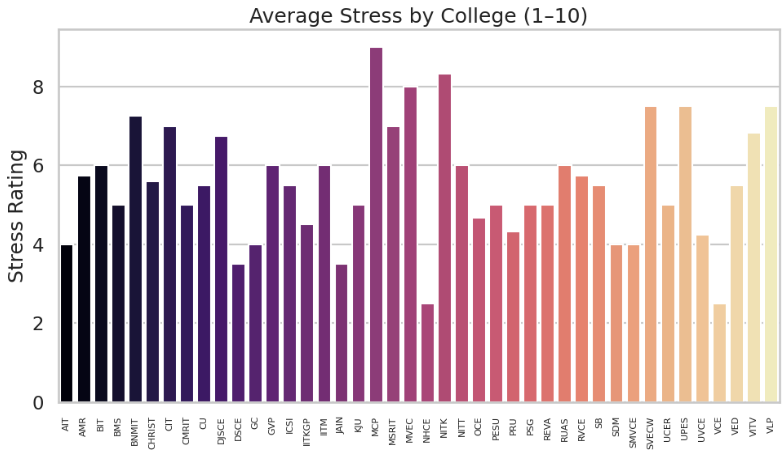


Observation: This comparative boxplot identifies how the use of specific campus zones influences emotional states. Students who "Use" Sports Grounds and Green / Open Areas report significantly higher median mood ratings, reaching approximately 8 out of 10, compared to those who do not use these spaces. Conversely, students who spend time primarily in Academic Blocks and Libraries show more variable mood distributions with lower medians.

Insight: These findings pinpoint exactly where well-being is generated on campus. While libraries and academic blocks are functional for study, they appear to be emotionally neutral or even taxing. The "Mood Boost" found in sports and green areas suggests that physical activity and nature exposure are the two most effective spatial interventions for improving the student psyche.

Stress Analysis

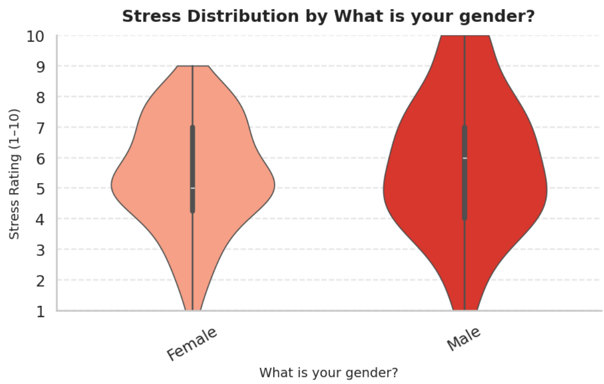
1. Average Stress by College

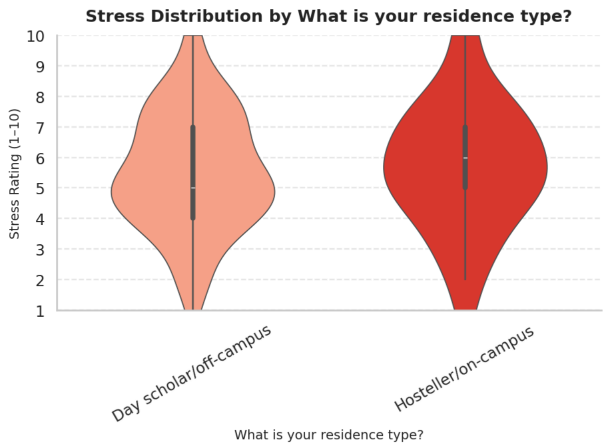


Observation: This chart displays a significant variance in stress ratings across different institutions, with scores ranging from approximately 2 to 9.

Insight: It identifies specific campuses as high-stress environments versus those that maintain notably lower stress levels, suggesting that institutional culture and campus design play a role in student well-being

2.Stress by Gender & Residence Type

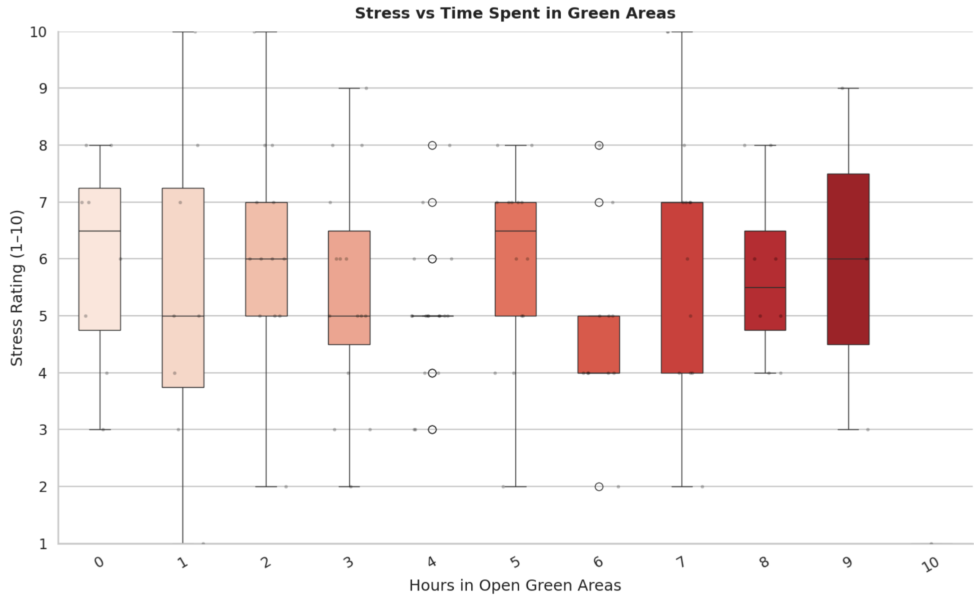




Observation: These plots compare stress distributions between genders and between day scholars versus hostel/on-campus residents.

Insight: While gender differences are minimal, male and hostel-based students exhibit marginally higher and more variable stress levels. This suggests that living in dense, urbanized campus environments (hostels) can increase psychological pressure.

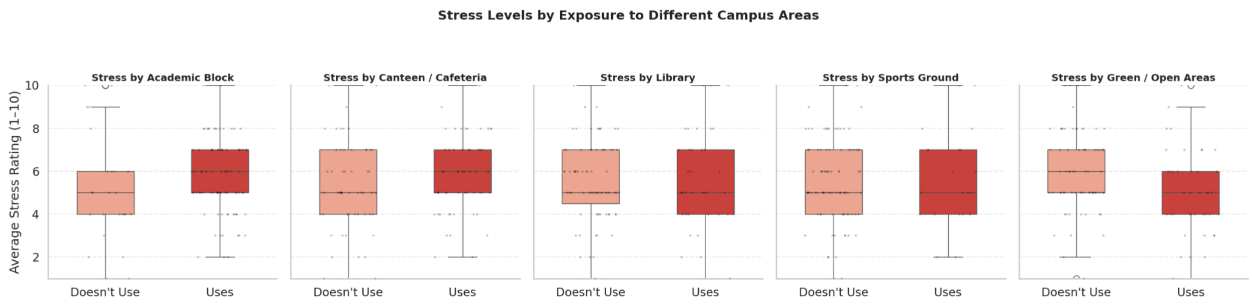
3. Stress vs. Time Spent in Green Areas



Observation: This chart tracks how the number of hours spent outdoors impacts stress ratings on a scale of 1–10.

Insight: There is a clear "tightening" of stress distributions as outdoor exposure increases. As students spend more time in green zones, the median stress level drops and the range of extreme stress responses narrows, confirming that nature acts as a stabilizing buffer against academic pressure.

4.Environmental Influence on Well-being

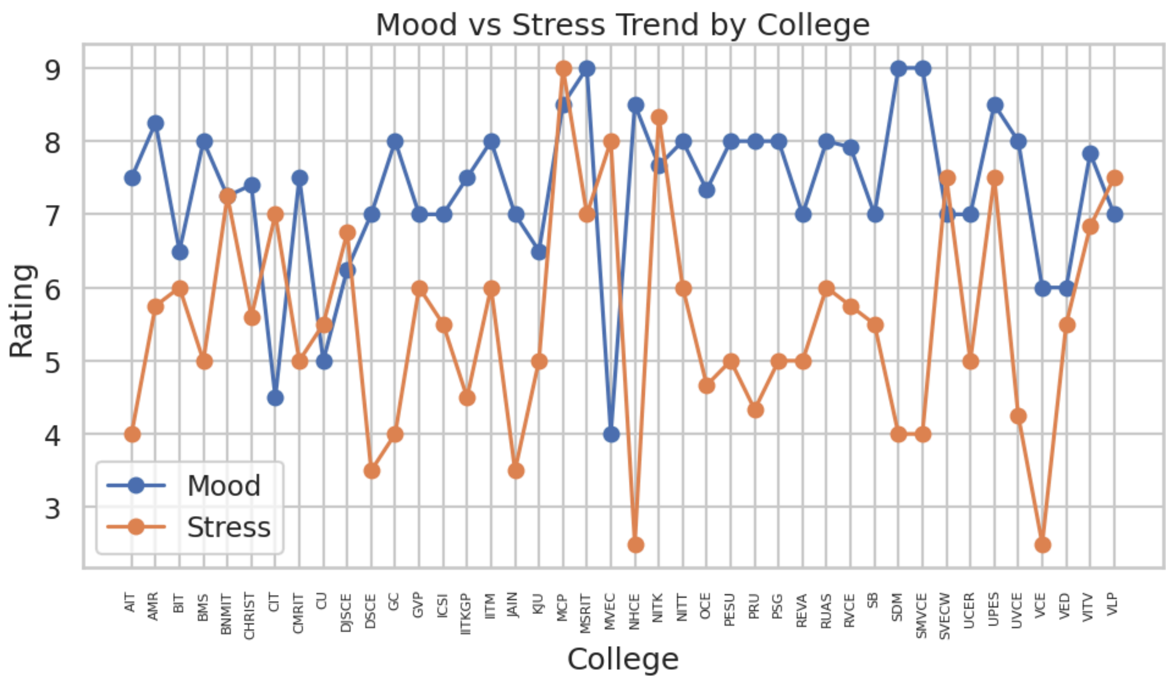


Observation: The box plots reveal a distinct divergence in stress response between "Academic" and "Green" zones; while exposure to Academic Blocks shifts the entire stress distribution upward, usage of Green/Open Areas is the only variable that trends the median stress level downward.

Insight: This identifies a clear "Environmental Buffer" effect. It suggests that while academic settings are inherent stress-drivers, natural landscapes on campus act as a functional intervention, successfully lowering the baseline stress of the students who frequent them.

HOLISTIC ANALYSIS:

1. Mood vs. Stress Trend by College



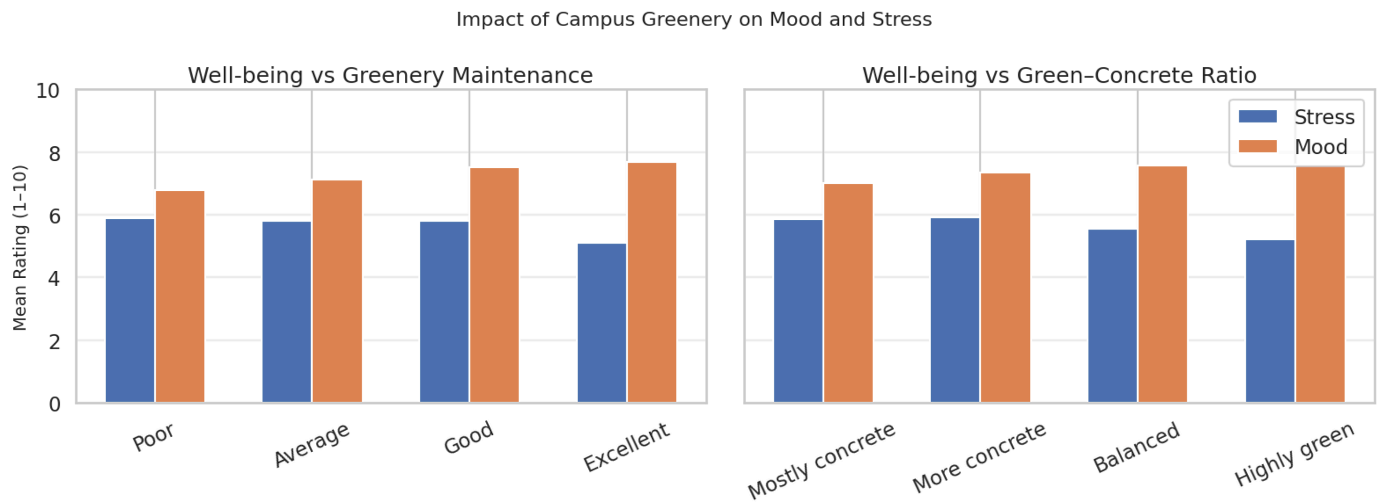
Observation: The dual line graph shows a consistent mirroring effect across various institutions; as the blue "Mood" line rises, the orange "Stress" line typically falls.

Insight: This highlights a healthy inverse relationship between well-being and academic pressure, suggesting that colleges successfully fostering positive emotional states naturally see a reduction in student stress.

2.Impact of Campus Greenery on Mood and Stress

Observation: The "Green-Concrete Ratio" bar chart shows that "Highly Green" campuses maintain lower stress levels and higher mood ratings compared to "Mostly Concrete" campuses.

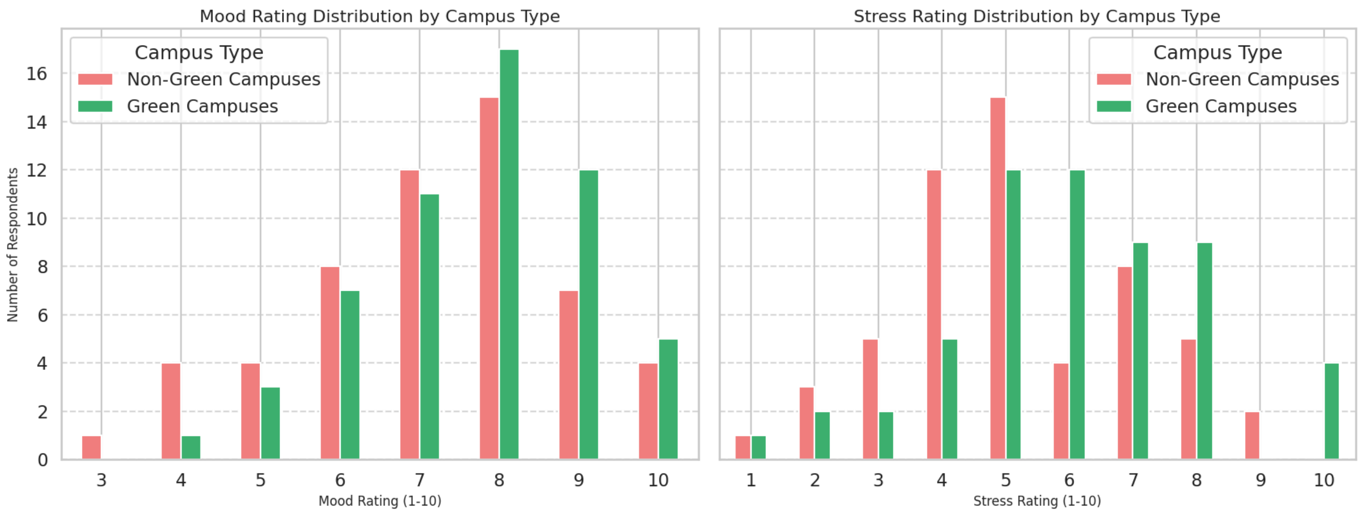
Insight: The physical composition of the campus is a primary determinant of mental health; an environment dominated by gray infrastructure (concrete) actively limits the restorative capacity of the student body.



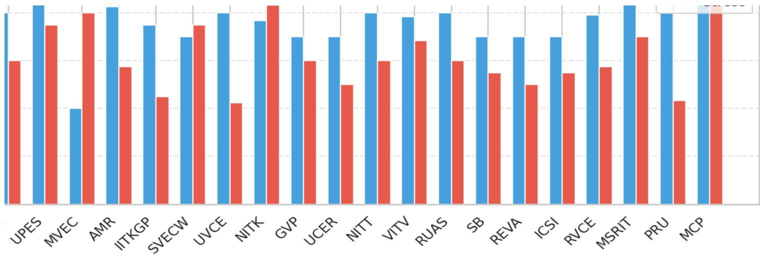
3. Mood Rating Distribution

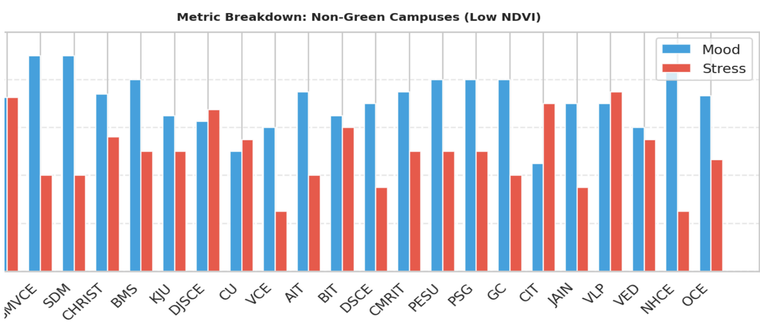
Observation: The histogram for green campuses is "right-skewed," peaking at a rating of 8, while non-green campuses peak much lower at 4–5.

Insight: Nature "lifts the floor" of well-being. It ensures that the "average" student on a green campus feels as good as the "top-tier" student on a concrete-heavy campus.



4.Well-being Comparison: Green vs. Non-Green Campuses

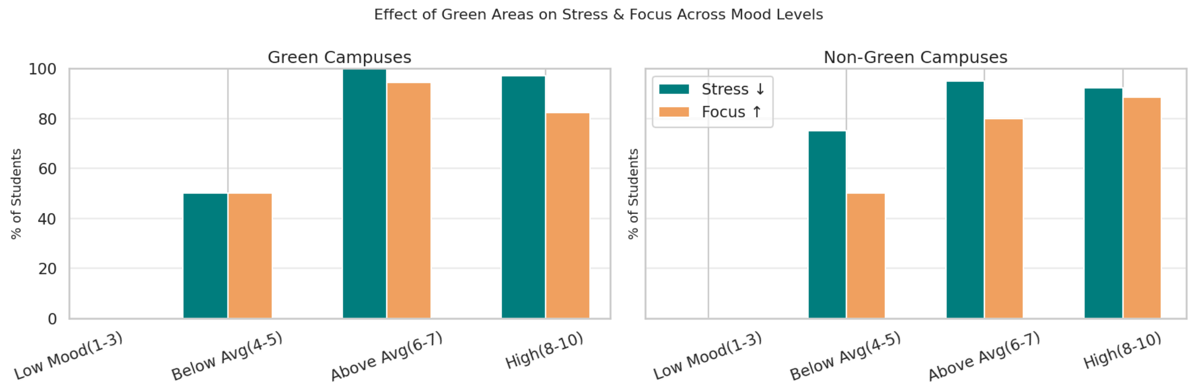




Observation: Green campuses (High NDVI) exhibit significantly higher average mood and noticeably lower average stress compared to non-green campuses (Low NDVI).

Insight: Green campuses possess a distinct "Well-being Advantage," confirming that environmental design is a critical pillar for supporting the student psyche.

5. Stress Rating Distribution



observation: Non-green campuses show a high concentration of students stuck at a constant stress level of **5**, whereas green campuses show a more distributed range.

Insight: Students in concrete environments suffer from "Chronic Mid-level Stress," never truly finding a way to decompress. Green environments allow for a "recovery cycle," preventing students from plateauing at high stress levels.

Key Outputs

- Cleaned datasets (survey + geospatial)
- GreenScore for each campus
- Visual dashboards: Green cover vs stress, Mood vs exposure to green spaces

Product Concept: Green Mind Metrics

Digital platform for:

- Campus GreenScore comparisons: High-stress, low-green campuses
- Green oasis finder within a campus: High-green, high-well-being zones within any campus (“green oases”)

Syllabus Alignment

- **Unit I (Data Understanding):** Handling structured (CSV) and unstructured (Satellite images) data; normalization and cleaning.
- **Unit II & III (Collection):** Primary survey design (Psychometric) and Sensor/Satellite logging (Geospatial pipeline).
- **Unit IV (Storytelling):** Creation of the Flask Dashboard, Scrollytelling narrative, and visual reports.
- **Unit V (Statistics & Ethics):** Correlation analysis, avoiding bias in survey interpretation, and ensuring data privacy (anonymized student data).

Expected Impact

- Helps students choose healthier study environments
- Helps campuses prioritize green investments
- Supports evidence-based campus planning
- Raises awareness of environment–mental health links
- Insights for planners and wellness centers

Contribution Table:

Student Name	**USN / Roll No.**	**Roles & Responsibilities**	**Contribution (%)**	**100% Active (Yes/No)**
A	1RV22CS089	Survey design & data collection	25	Y
B	1RV22CS099	Cleaning & analysis	25	Y
C	1RV22CS105	Product ideation & mockup	25	Y
D	1RV22CS137	Report & storytelling	25	Y